



ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

SEMESTER 1 (2024/2025)

CCC2143

SOFTWARE ENGINEERING

GROUP PROJECT

“AIU ALUMNI MANAGEMENT SYSTEM”

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Submission Date : 17.1.2025

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1. Introduction

The Alumni Management System is an innovative platform designed to bridge the gap between students and alumni, fostering collaboration, mentorship, and professional growth. By leveraging modern technologies and employing the Integration and Configuration software process model, this system serves as a centralized hub for networking, and knowledge-sharing. Students can explore alumni profiles, engage in discussions, and access curated opportunities such as job, educational postings and events. Alumni, on the other hand, can offer insights ,create and view jobs,educational postings and contribute to the development of the university community. The platform also empowers administrators with tools to manage content, oversee registrations, and ensure seamless operation. This initiative aligns with Albukhary International University's mission to strengthen its community by fostering enduring connections and supporting career advancement through shared knowledge and opportunities.

PROJECT DESCRIPTION	This project is about developing a specific web platform that will facilitate communication between Albukhary International University's current and former students. The website will act as a venue for deep communication, encouraging knowledge exchange and mentoring. In order to assist students in making well-informed decisions on their future, alumni will provide information on employment opportunities, career guidance, and share personal experiences. Alumni from a variety of businesses can provide current students with direct access to a range of viewpoints and useful insights. By creating a friendly and cooperative community, the platform will help improve the university's network. In order to motivate and empower users, it will include discussion boards, job openings, mentorship possibilities, and success stories. Through mutual assistance and pooled resources,this platform will serve as a transformative resource, empowering AIU students and alumni to achieve sustainable growth and professional success.
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PROBLEM STATEMENT	Albukhary International University (AIU) lacks a dedicated digital platform to connect current students with its alumni, particularly for career-related opportunities. While tools like Telegram are used for communication, they are not tailored to address specific needs such as job placements, mentorship, or advanced study guidance. This limitation creates a gap in leveraging the university's alumni network to provide valuable career insights, professional development opportunities, and access to job markets. Without a centralized and purpose-built platform, students face challenges in making informed decisions about their future careers or pursuing master's programs. Students find it difficult to obtain open internships and employment opportunities. This project aims to address these challenges by developing a specialized web platform that bridges this gap, providing information about available opportunities, fostering collaboration and knowledge exchange.
PROJECT OBJECTIVES	1. Create a robust connection between students and alumni at Albukhary International University. 2. Register at least 30% of AIU alumni and 40% of current students within the

	first year. 3. Facilitate 40 mentorship connections between alumni and students within six months. 4. Host and promote 10 alumni-led job or internship postings every quarter. 5. Encourage the creation of 30 career-oriented discussion threads within the first year. 6. Publish 20 alumni success stories within the first year to inspire users. 7. Improve employment outcomes for AIU graduates by 15% within two years.
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2. Gantt Chart

The Gantt chart of the project is attached in the appendix (1).

3. System Requirements Analysis

3.1 User Requirements

Students:

- Register as Student
- Log in as student
- View alumni profiles for networking and guidance.
- View success stories, job postings, educational opportunities, and upcoming events.
- Create and comment on discussion forums

Alumni:

- Register as Alumni
- Log in as Alumni
 - Update and manage their profile.
 - View profiles of other alumni.
 - Post job opportunities
- Post Educational opportunities.
 - View success stories, job postings, educational opportunities, and upcoming events.
 - Create and comment in discussion forums to engage with students and other alumni.

Admi:

- Verify and approve alumni and student registrations.
- Manage job postings, educational opportunities, success stories, and upcoming events.
- Moderate and manage discussion forums.
- Announce new events or updates to engage users.

3.2 System Requirements

Functional Requirements

- **Authentication:** Users (alumni, students, admins) should not enter the system without before they provide the correct access codes.
- **User Management:** Registration concerning alumni and students and assistance in management of their profiles.
- **Admin Controls:** Registrations need to be approved, and content managed, therefore admins should be allowed access.
- **Content Management:** This will enable job posts, educational opportunities, success stories, and events to be posted, accessed, edited and removed.
- **Search and Filter:** Allowing searching and filtering of alumni profiles, jobs and educational opportunities.

- **Discussion Forum:** An open forum where alumni can have real-time discourse with students.

Non-Functional Requirements

- **Requirements Scalability:** The system should be able to accommodate a possibility of increased user registrations and more data in the future.
- **Performance:** Make sure to offer fast feedback time on the actions the users perform on the website such as login, search and moving from one page to another.
- **Security:** Make safe access and federation from different identities and applied security measures to protect data from leaks.
- **Usability:** Facilitate easy to use and smooth usability where users can navigate frequently and interact without difficulty.
- **Accessibility:** The website should be usable by disabled persons (for example, a web-site has to be readable by a post-visitor screen-reader application).

3.3 Software Process Model: Integration and Configuration

Integration and Configuration model is a software development paradigm that involves integrating and configuring existing software components or systems to a project need. This model enables developers to tap from existing pre-built solutions hence reducing the time and costs as well as guarantee good and proven results (Patil, 2022). With regard to this project, the model will solicit for an existing web platform code base and adapt it to cater for the alumni-student portal in Albukhary International University.

The Integration and Configuration Model is chosen for its efficiency, flexibility and ability to reuse components for tailored, cost-effective solutions.

- **Time Efficiency:** Due to the use of such parts, the rate of development is increased and the team can use most of the time to tailor the design to suit its needs instead of implementing standard elements. (Shafiee, et al., 2022)

- **Cost-Effectiveness:** Pre-built systems are less expensive to acquire than developing a new system from scratch, therefore control over the design and code of each feature individually is eliminated.
- **Reliability:** The consequent results include pre-tested various parts of the program that take less bugs and issues so the platform becomes stable. (Bendix, n.d.)
- **Customization Potential:** Current deployed systems are an effective base to start with while setting up options to change and add-on features according to the project's needs.

Application in This Project

1. Sourcing the Codebase:

The project team sourced an open-source or pre-built web application whose structure closely resembled the required platform. Examples considered included content management systems such as WordPress and open-source educational platforms.

2. Customization of Features:

- **Forums:** A clear and easy-to-navigate forum for career-related topics was created. Ban and muffle features were added to ensure appropriate conversations among students.
- **Alumni Success Stories:** A section was designed to showcase alumni and students' achievements and milestones..
- **Job Postings and Opportunities:** A module for posting internships, employment opportunities, and studies for alumni was developed. Search filters and application links were interlinked for ease of use.

3. User Interface Enhancements:

- The interface was customized to align with the university's branding, including its logo, colors, and identity.
- The platform was designed to be user-friendly, ensuring ease of navigation and accessibility across all devices.

4. Integration:

The constructed modules were integrated into a unified and smooth system, tailored to meet the specifics of various users.

5. Testing and Deployment:

Thorough testing was conducted to prevent conflicts between modules when working in conjunction. Both systems were stabilized, and the fully developed system was hosted on servers accessible to AIU students and alumni.

4. System Model

4.1 Use Case Diagram

The use case diagram of the system is attached in the appendix (2).

4.2 Activity Diagram

The activity diagram of the system is attached in the appendix (3).

4.3 Sequence Diagram

The sequence diagram of the system is attached in the appendix (4).

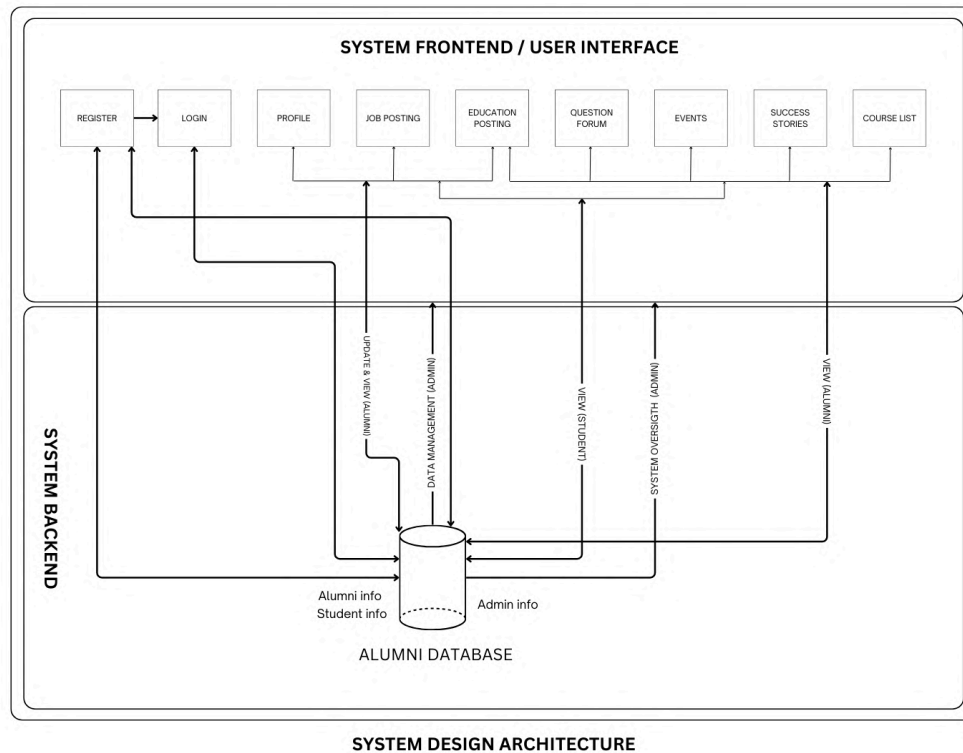
4.4 State Diagram

The State diagram of the system is attached in the appendix (5).

5. System design Architecture

Overview:

The Alumni Management System consists of a System Frontend/User Interface and a System Backend. The frontend allows users (alumni, students, and administrators) to interact with the system, while the backend processes and stores data in dedicated databases for efficient management.



5.1 System components:

The interface offers several modules for interaction:

- Register/Login: Enables user authentication and access control.
- Profile: Allows users to view and update their information.
- Job Posting: Facilitates alumni to share job opportunities with students.
- Education Posting: Enables sharing of educational resources and opportunities.
- Question Forum: A platform for alumni and students to ask and answer questions.
- Events: Displays upcoming alumni and university events.
- Course List: Provides a directory of available university courses, including descriptions and enrollment information.
- Success Stories: Highlights achievements of alumni and students.

5.2 System Backend:

The backend consists of three primary databases and the admin management system:

Alumni Database: Stores and manages alumni information.

- Data is retrieved for profile views, job postings, and events.
- Alumni can update their data through the frontend.
- Maintains student details: Accessed for event participation, education postings, and question forums.
- Admins manage the system by overseeing data integrity, updating content, and maintaining access controls.

5.3 Data Flow and Interaction:

- **Frontend to Backend:**

- Users (alumni or students) interact with various modules on the frontend, sending requests to retrieve or update data in their respective databases.
- Admins interact with the admin database to ensure smooth system operation and data consistency.
- Technologies: HTML, CSS, JavaScript.

- **Backend to Frontend:**

- The backend processes user requests, retrieves or updates information in the appropriate database, and delivers responses to the frontend for display.
- Technologies: Python frameworks such as Flask or Django, or FastAPI for robust API development.

5.4 Key Features and Functions:

- **User Management:** Offers secure registration and login for all users, Implements password encryption and authentication measures.
- **Data Viewing and Updating:** Allows users to access and update their profiles, job postings, educational resources, and event details in real time.
- **Collaboration Tools:** Includes forums, job boards, and event notifications to foster alumni-student engagement.

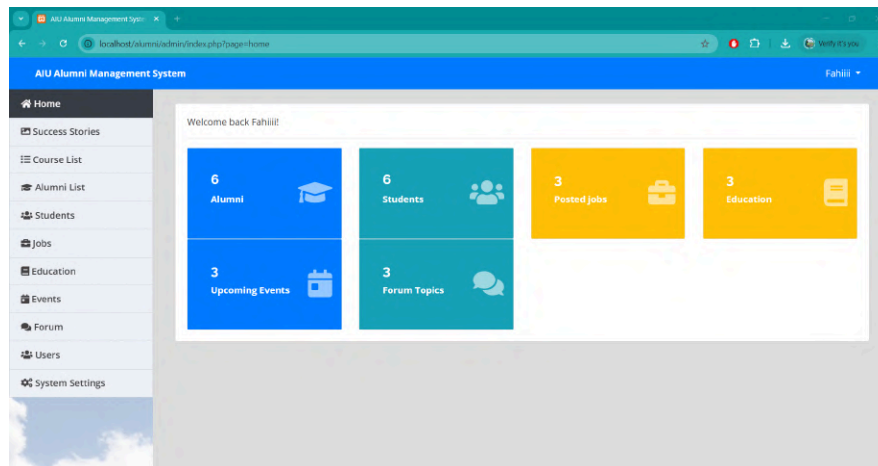
- **Admin Oversight:** Administrators maintain data consistency, system functionality, and manage access rights efficiently.

5.5 Technical Considerations:

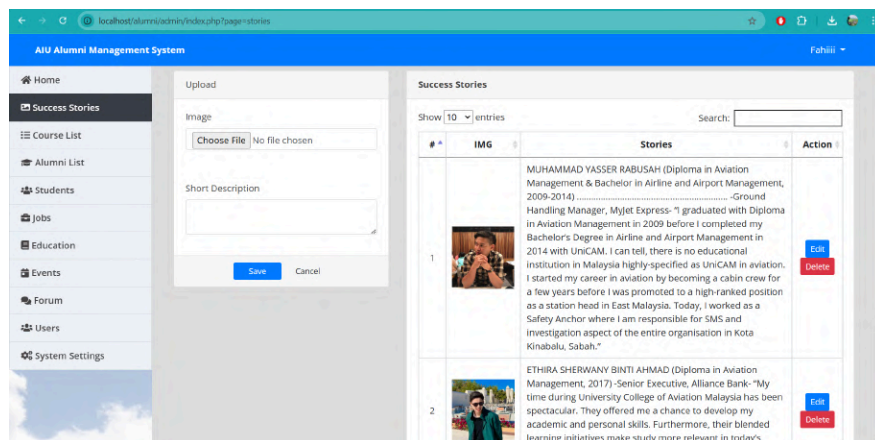
- **Scalability:** The system should handle increasing alumni and student data as the user base grows.
- **Security:** Implements robust authentication protocols, data encryption, and secure database connections to prevent unauthorized access also Secure communication via HTTPS.
- **Integration:** Ensures smooth communication between frontend modules and backend services through well-defined APIs.
- **Performance:** Optimized database queries, caching mechanisms, and responsive user interfaces ensure fast and reliable performance.

6. Implementation of Prototype System

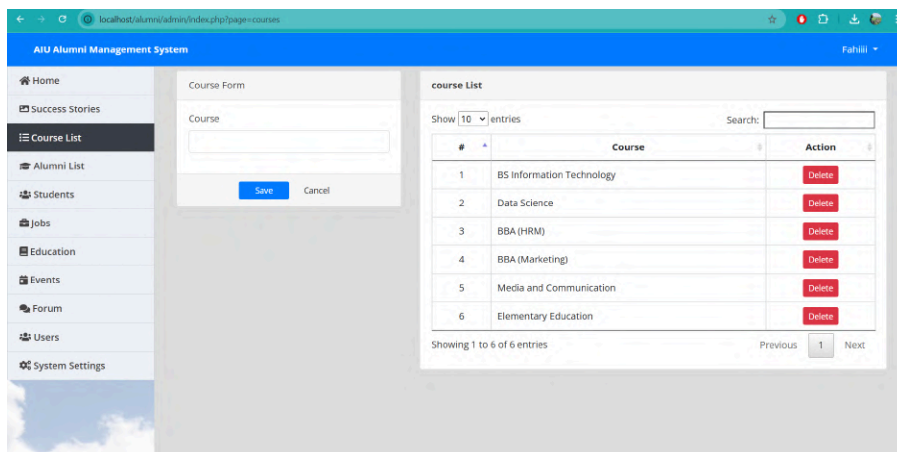
The AIU alumni management system was developed and implemented using different programming languages for frontend and backend. The languages used are HTML, CSS, JQuery, Bootstrap and PHP/MySQL.



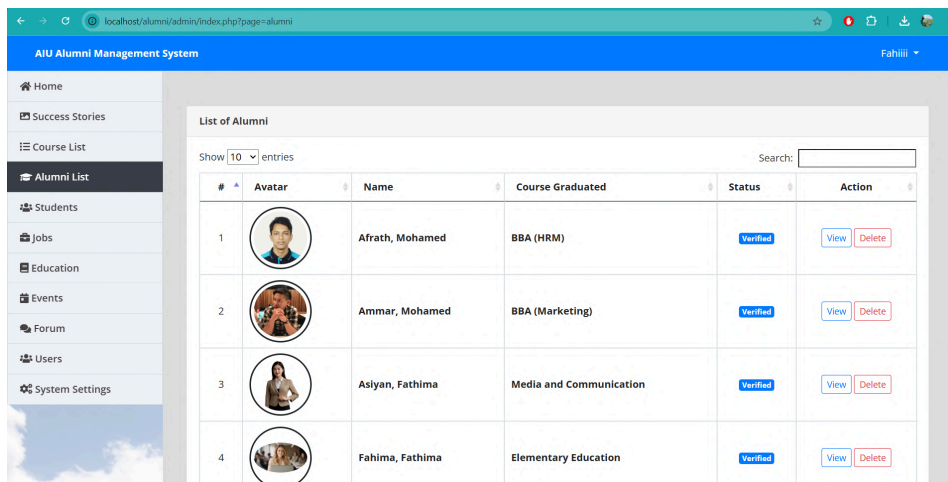
Admin side dashboard



Manage success stories



Manage Courses list



Manage Alumni

localhost/alumni/admin/index.php?page=student

AIU Alumni Management System Fahili

Home Success Stories Course List Alumni List **Students** Jobs Education Events Forum Users System Settings

List of Students

Show 10 entries Search:

#	Name	Course	Status	Action
1	Anshaf, Salam	Data Science	Verified	View Delete
2	Asar, Khan	BS Information Technology	Verified	View Delete
3	Faruk, Mohamed	Data Science	Verified	View Delete
4	Newer New, New	BS Information Technology	Verified	View Delete
5	Ridwan, Af	Media and Communication	Verified	View Delete
6	Safna, Fathima	BBA (HRM)	Verified	View Delete

Showing 1 to 6 of 6 entries Previous 1 Next

localhost/alumni/admin/index.php?page=home

Manage Students

localhost/alumni/admin/index.php?page=jobs

AIU Alumni Management System Fahili

Home Success Stories Course List Alumni List Students **Jobs** Education Events Forum Users System Settings

Jobs List

Show 10 entries Search:

#	Company	Job Title	Link	Posted By	Action
1	Meta	Digital Marketing Specialist	https://www.metacareers.com/locations/malaysia/?p(offices)[0]=Kuala%20Lumpur%2C%20Malaysia&offices[0]=Kuala%20Lumpur%2C%20Malaysia	Admin	View Edit Delete
2	MCMC	Cyber Security Executive	https://my.jobstreet.com/cybersecurity-jobs?jobid=81157776&type=standout	Mohamed Afrath	View Edit Delete
3	ABC Company	IT Specialist	No Link	Admin	View Edit Delete

Showing 1 to 3 of 3 entries Previous 1 Next

Manage Job Postings

localhost/alumni/admin/index.php?page=education

AIU Alumni Management System Fahili

Home Success Stories Course List Alumni List Students Jobs **Education** Events Forum Users System Settings

Education List

Show 10 entries Search:

#	Institution	Program	Posted By	Link	Action
1	Google (Offered Via Coursera)	Google IT Support Professional Certificate	Admin	https://grow.google/certificates/it-support/	View Edit Delete
2	Harvard University	Online - Data Science Professional Certificate	Fahili	https://www.harvard.edu/	View Edit Delete
3	German Academic Exchange Service (DAAD)	Master's Degree Programs in Germany	Admin	https://www.daad.de/en/studying-in-germany/scholarships/daad-scholarships/	View Edit Delete

Showing 1 to 3 of 3 entries Previous 1 Next

Manage Educational Postings

AIU Alumni Management System

Home Success Stories Course List Alumni List Students Jobs Education **Events** Forum Users System Settings

List of Events [+ New Entry](#)

Show 10 entries Search:

#	Schedule	Title	Description	Committed To Participate	Action
1	Feb 14, 2025 06:30 AM	Alumni Get-together 2025	The Alumni Get-together is organized on Saturday, February 14, 2025 @ 6:30 PM. The Century Club, Bangladesha Bangladesha will be the venue to re-ite...	1	View Edit Delete
2	Jan 22, 2025 08:00 PM	Study Circle January	Study Circle for SC with Fahima	3	View Edit Delete
3	Jan 11, 2025 11:00 PM	Skill Development Lecture Series	The Skill Development Lecture Series is a curated program designed to empower individuals by enhancing their personal and professional...	2	View Edit Delete

Showing 1 to 3 of 3 entries Previous 1 Next

Manage Events

AIU Alumni Management System

Home Success Stories Course List Alumni List Students Jobs Education Events **Forum** Users System Settings

Forum List [+ New](#)

Show 10 entries Search:

#	Topic	Description	Created By	Comments	Action
1	Skill Development And Lifelong Learning	Recommend courses, workshops, or certifications that have helped you professionally. Share your experiences on...	Mohamed Faruk	0	View Edit Delete
2	Career Guidance And Opportunities	Share advice, resources, and opportunities for career growth. Alumni can post about job openings, internships, or tips on succeeding in specific...	Admin	0	View Edit Delete
3	Data Science	Suggest some good python libraries that can be used for data analytics	Admin	0	View Edit Delete

Showing 1 to 3 of 3 entries Previous 1 Next

Manage Forums

User Side

Home

AIU Alumni Management System

Home Alumni Stories About Login as Alumni Login as Student

Welcome to AIU Alumni Management System

Upcoming Events

RANGE OF MOTION LECTURE SERIES

Skill Development Lecture Series
January 11, 2025 11:00 PM

The Skill Development Lecture Series is a curated program designed to empower individuals by enhancing their personal and professional competencies. Through a series of expert-led sessions, participants gain valuable insights, practical knowledge, and actionable strategies to excel in various fields. Key topics such as communication, de...

Alumni signup and login

AIU Alumni Management System

Login

Email

Password

[Create New Account](#)

Login

Welcome to AIU Alumni Management System

Upcoming Events

Range of Motion Lecture Series

Skill Development Lecture Series

January 11, 2025 11:00 PM

The Skill Development Lecture Series is a current program designed to empower individuals by enhancing their personal and professional competencies. Through a series of expert-led sessions, participants gain valuable insights, practical knowledge, and networking opportunities that will aid in their career growth and personal development.

AIU Alumni Management System

Home Alumni Stories About Login as Alumni Login as Student

Create Account

Last Name

First Name

Middle Name

Gender

Male

Batch

Course Graduated

Please Select Here

Current Company/Institution

Image (Picture of you)

Choose File No file chosen

Email

Password

Create Account

Upcoming Events

AIU Alumni Management System

Home Alumni Stories Jobs Education Forums About Mohamed Afrath

Study Circle January

January 22, 2025 08:00 PM

Study Circle for SCI with Fahima

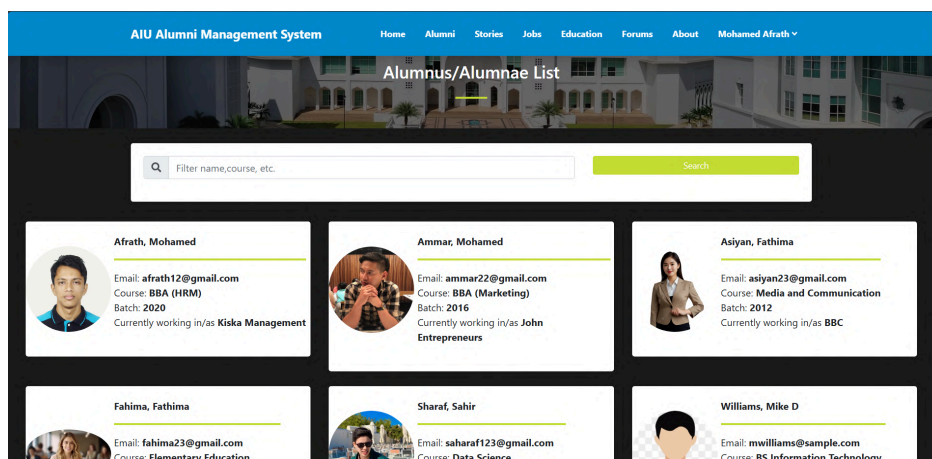
[Read More](#)

Alumni Get-together 2025

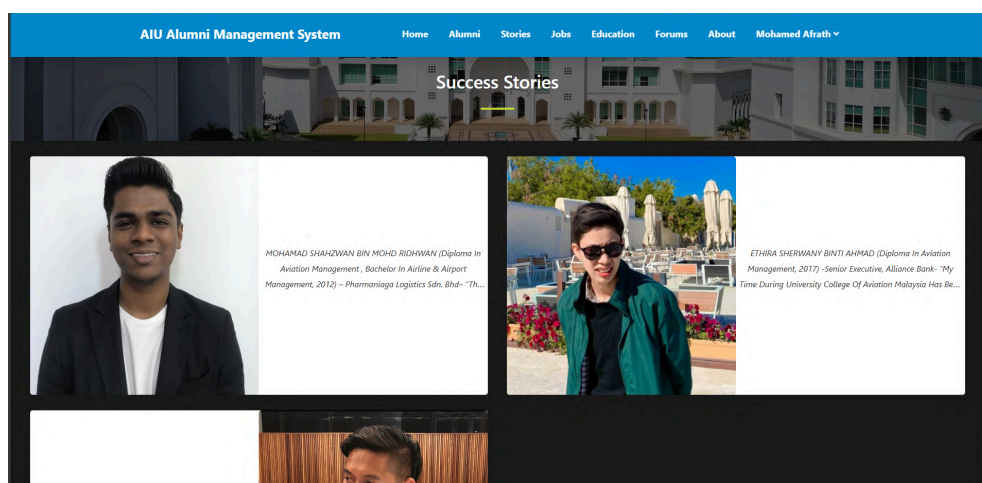
February 14, 2025 06:30 AM

The Alumni Get-together is organized on Saturday, February 14, 2025 @ 6:30 PM. The Century Club, Bengaluru will be the venue to re-live those happy moments with SD Miles. Please register to connect with your fellow alumni in this relaxing venue.

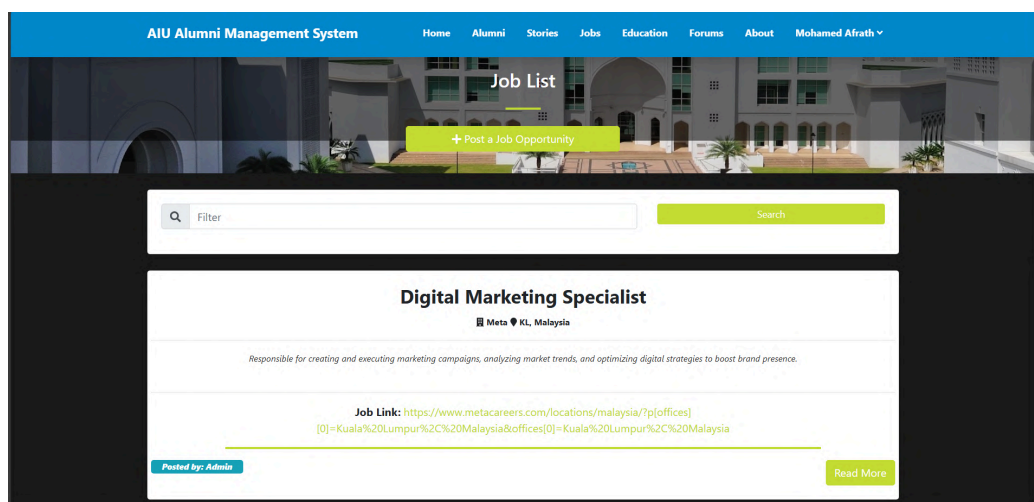
Alumni List Dashboard

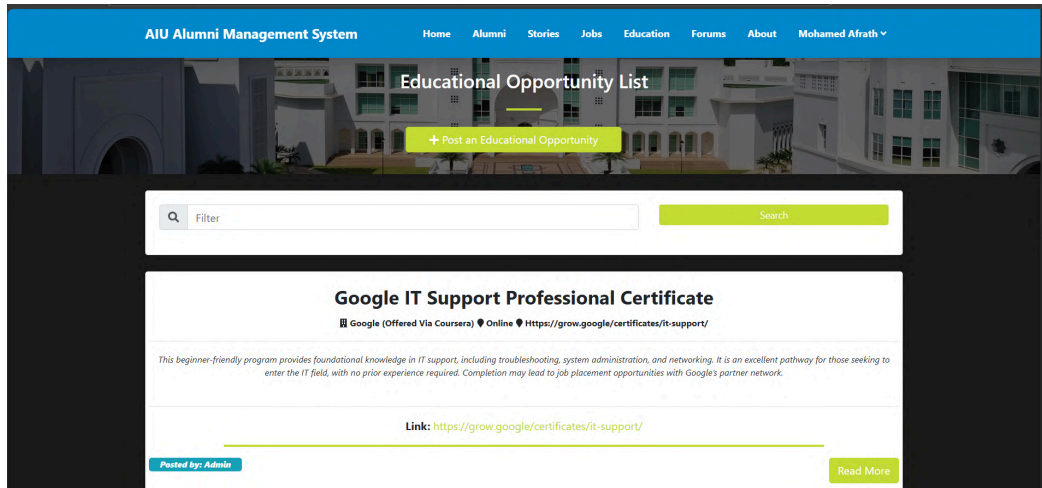


Success Stories Gallery

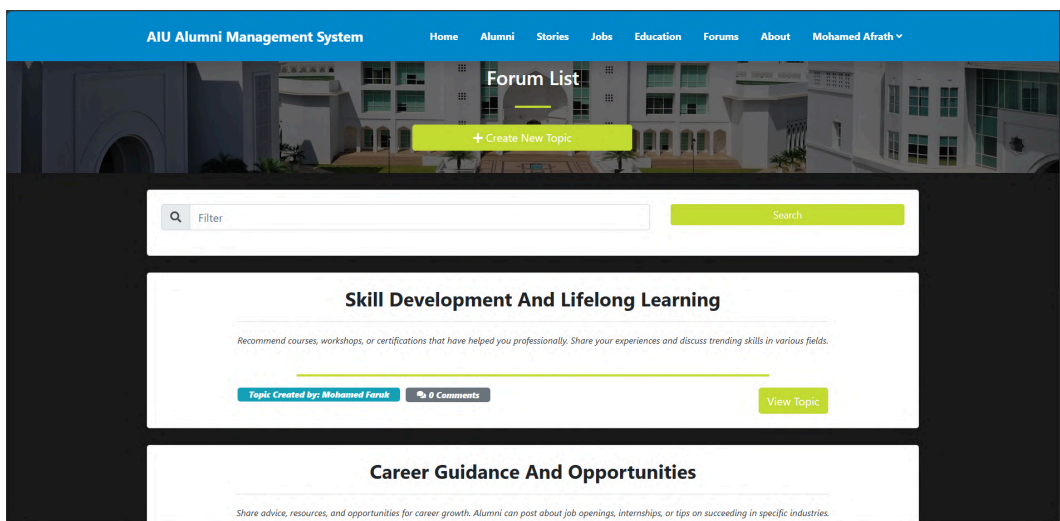


Jobs and Education Postings





Forums & Discussion Topics



7. System Testing Strategies

7.1 Development Testing

1. Unit Testing

Objective: Check specific functions or methods from the module by using sample inputs to validate if it works the way you want.

Test Cases:

- Login Module: Check if users can enter correct passwords, and their passwords are processed incorrectly; refuse entry with incorrect passwords and lock accounts.
- Profile Module: Ensure that the 'edit profile' feature enables the Alumni or Student details to be modified in the database.
- Posting Module: The forms of input are tested with functions such as "Create Port" and "Delete Port" in valid and invalid modes.

Testing Scenarios:

- See that while preparing the form, the fields which are mandatory fields such as name, email, etc., cause validation errors if left blank.
- Make sure that queries to the database such as a retrieval of user profiles yields the right information.

Metrics:

- Code coverage should be higher than 90% and it would play a major role in result testing since all or almost all lines of code and the paths through the program are covered.

2. Component Testing

Objective: Check how different modules share and exchange information and ensure that they do it correctly.

Test Scenarios:

- Database Integration: Shall check to see whether alumni job postings are properly stored and retrieved in the system. For instance, make sure the creation of a job post adds an entry in the database and looks good on the platform.
- Role Management: Confirm that Role Based Access Control works as expected, for example, alumni can post for jobs and students can only view.

Error Handling:

- Ensure that the system behaves well during and after failure such as a disconnected.

3. System Testing

Objective: Assess the conformity of the whole system to functional and nonfunctional necessities.

Functional Tests:

- Make sure that every job posted will be displayed on the dashboard once submitted.
- Check whether registering to the event changes ‘participation list’ for students and alumni in the database.
- Also integrate full sequences of tests, including account creation, event posting, and responses management.

Non-Functional Tests:

- Performance: Record latency times for those tasks that are likely to be executed by a large number of users at a single time, e.g. 50 users.
- Compatibility: Ensure the platform works well in various browsers incl. Chrome, Firefox and Edge as well as different devices incl. mobile, tablet, desktop.
- Scalability: Buy more users and data, to make sure that the system is ready to grow.

7.2. User Testing

Objective: Gather the feedback from actual users of the system in order to judge its functionality and efficacy.

Beta Testing:

- Collect data on mandatory functions such as site interface, content and flow.
- Test problems before the broad public gets a chance to use your software or website and report all the bugs and usability issues to you.

Usability Testing:

- See the users’ complete activities that include account registration, profile browsing and sharing jobs.

- Capture such parameters as finished tasks, the time required to accomplish tasks, and number of errors.

Accessibility Testing:

- Check for WCAG (Web Content Accessibility Guidelines) conformance –the standards for making web content more accessible.
- Check how your application works with screen reader and in keyboard navigation to ensure the graphical interface is partially usable for the visually impaired.
- Also, pay careful attention to ensure that all text and images of text have a contrast between them and their background and all fonts are large enough to read easily.

Surveys and Feedback Forms:

- Administer post-testing questionnaires for collecting open-ended responses about the ease of use of resources, satisfaction, and changes for the better.

8. Challenges and Opportunities

8.1 Challenges:

- Engaging alumni for them to contribute to the platform, by providing content.
- Continuing their interaction with users through timely and actual updates and notifications.
- Securing data for information regarding the users and their communication.
- Catering to various users – Employers for posting jobs, seekers for finding mentors, and everyone for forums.

8.2 Opportunities:

- They could have leveraged an existing social media site to build the community but instead, the AIU opted to develop its own concrete, focused site.
- Providing proper interaction between the student and alumni for career development.

- Strengthening AIU image with success stories or even better employment destiny of its students.
- Creating indispensable facilities that may expand as the user requirements emerge.

9. Role of Members

1. **Afrath (Front-end Developer):** Is fully accountable for the concept and development of user interfaces, maintaining continuity of the users' experience and adaptation of the site across all platforms: from mobile to personal computer.

2. **Faruk (Front-end Developer):** Jointly optimized UI components with Afrath, collaborates on the creation of adaptable designs, and handles front-end to back-end connectivity.

3. **Anshath (Back-end Developer):** Develops the server-side aspects consolidating the applications and organizing the data management system with secure/server oriented design. Is involved in the execution of such features as user identification, jobs uploading, and mentorship tools.

4. **Zin May Oo (Back-end Developer):** Works with Anshath on data managing, and on how the components should be integrated. Takes care of system optimization and also the way that the system is expanded to fit a large amount of data.

5. **Asiyan (Report and Documentation):** Responsible for project documentation such as systems requirements, project progress reports and other project documentation. Academics provides oversight on how best to document all the details of a given project as well as how to present them.

10. Conclusion

The notion of the platform for Albukhary International University will enhance communication between students and alumni in order to create a productive cooperation and successful job search. Through features of job listing, discussion boards and success stories, it allows students to get professional assistance and relevant connections while the alumni receive a platform in which they can provide constructive insights and opportunities. Based on the

Integration and Configuration software process model the platform will be developed with high reliability and scalability and leaning on the existing systems. This initiative helps to expand AIU's community, improve the professional result, and correlate with AIU's mission to support students and alumni by the conjoined efforts of knowledge-sharing.

11. References

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https://www.sciencedirect.com/science/article/pii/S2212827122003821?ref=cra_js_challenge&fr=RR-1

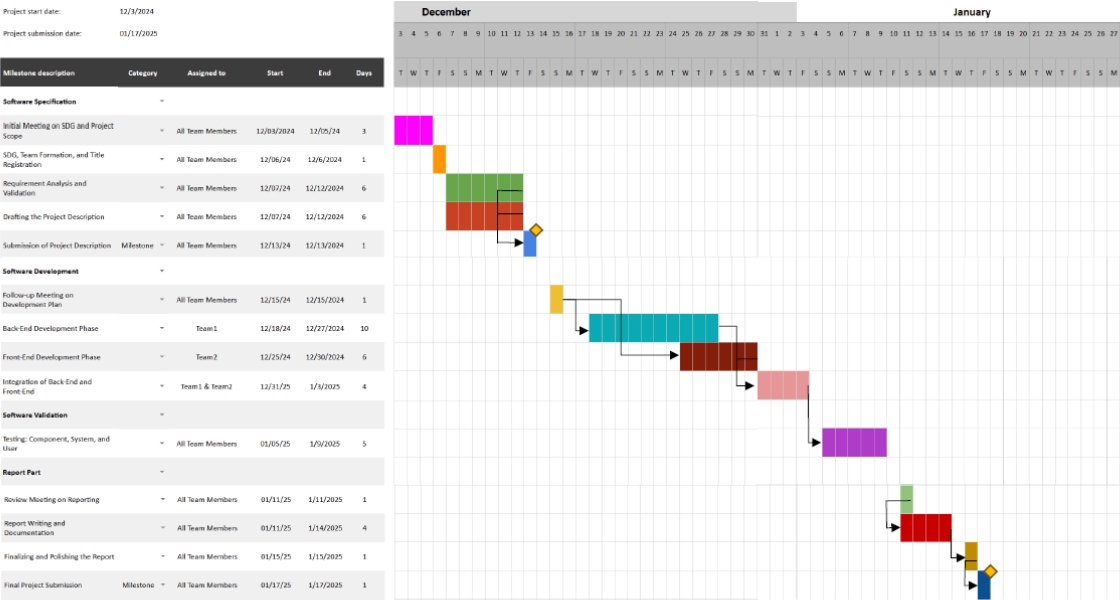
12. Appendix

https://drive.google.com/drive/folders/1Ls6zc_Lb9rpNLBm1HyaTnlsSEmDSSxAb

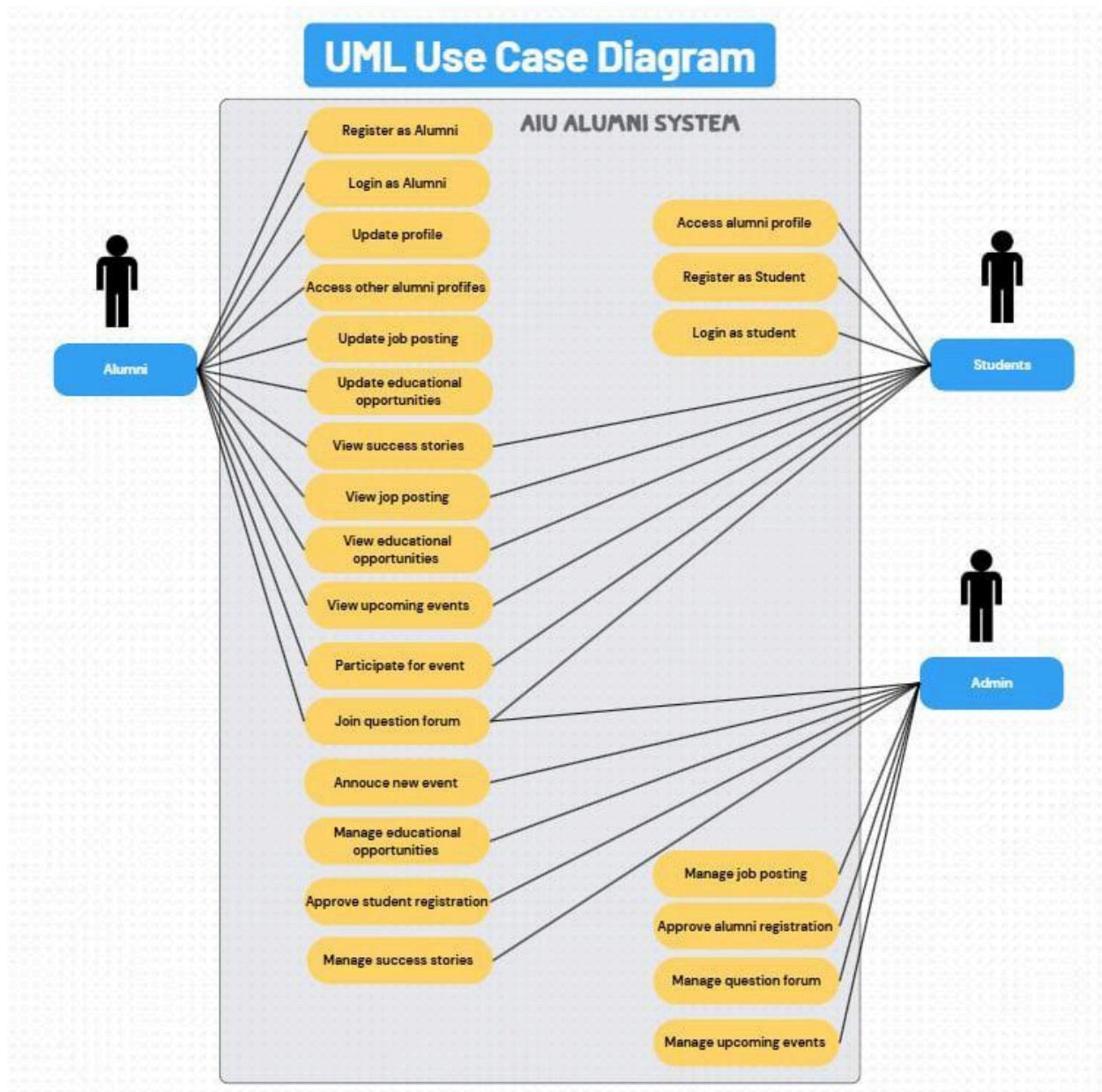
Link to project zip file

(1) Gantt Chart

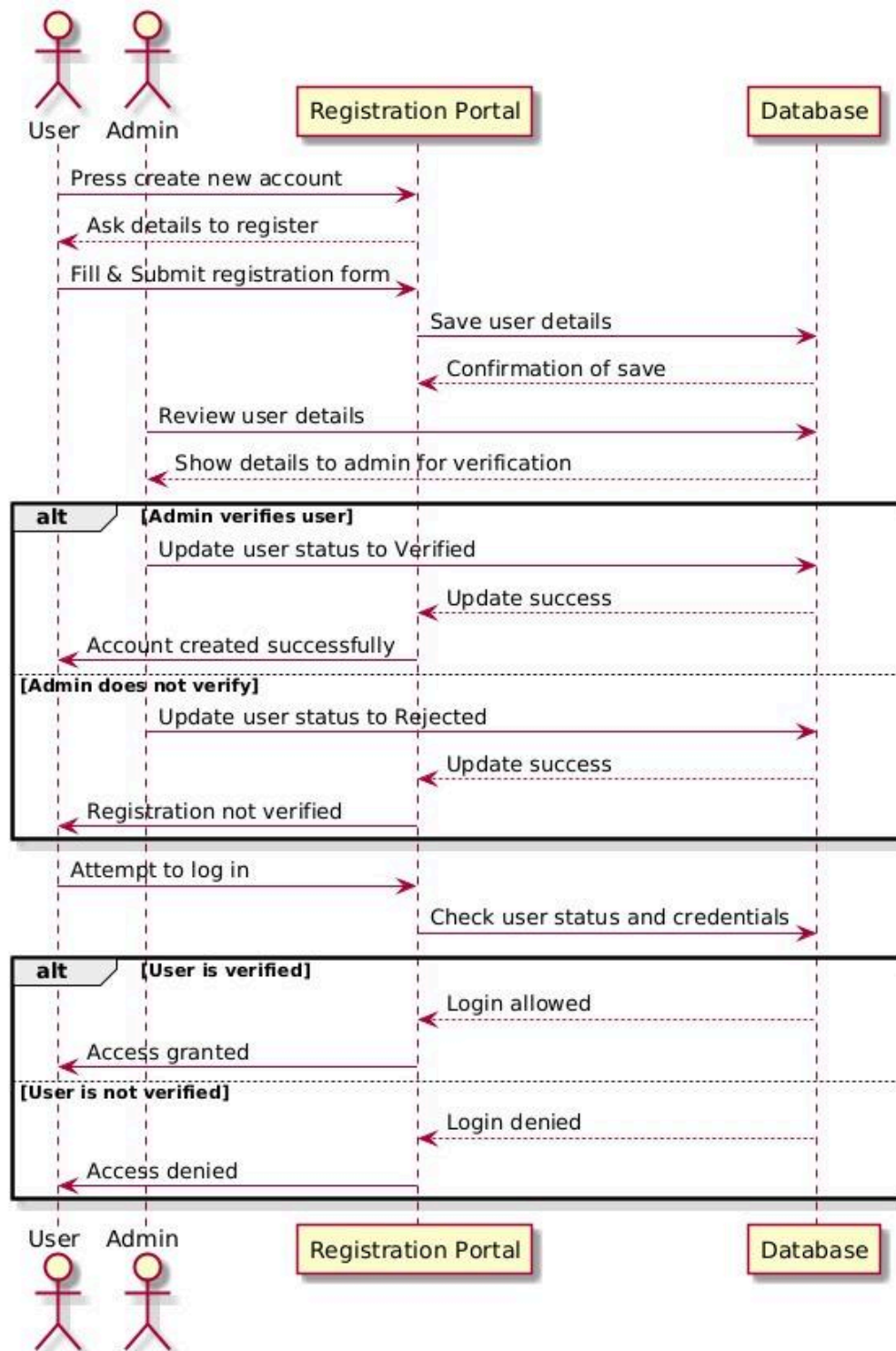
Project: A Digital Platform Connecting AIU Students with AIU Alumni



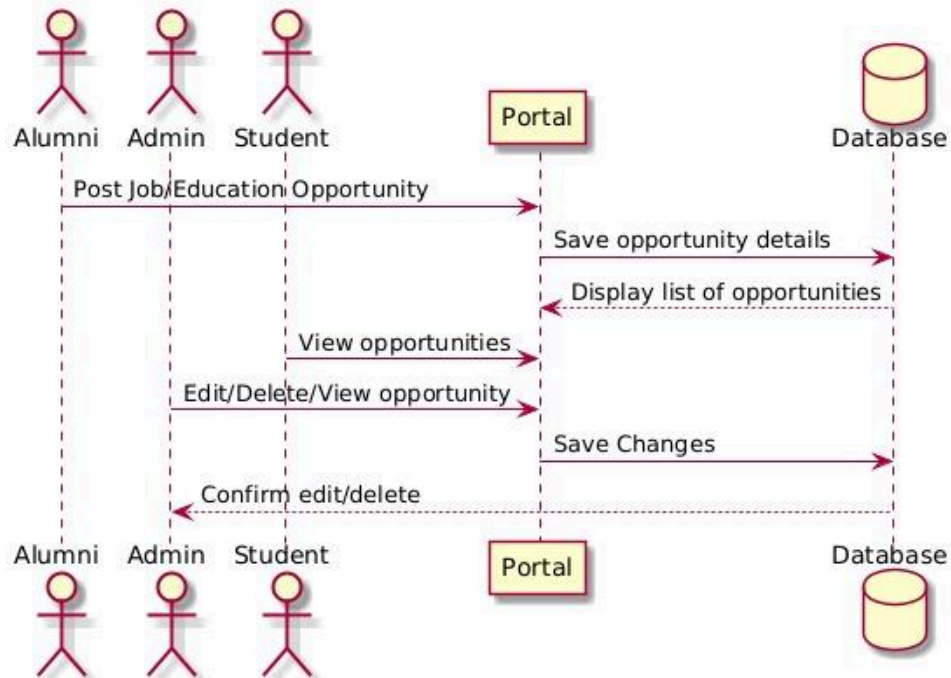
(2) Use Case Diagram



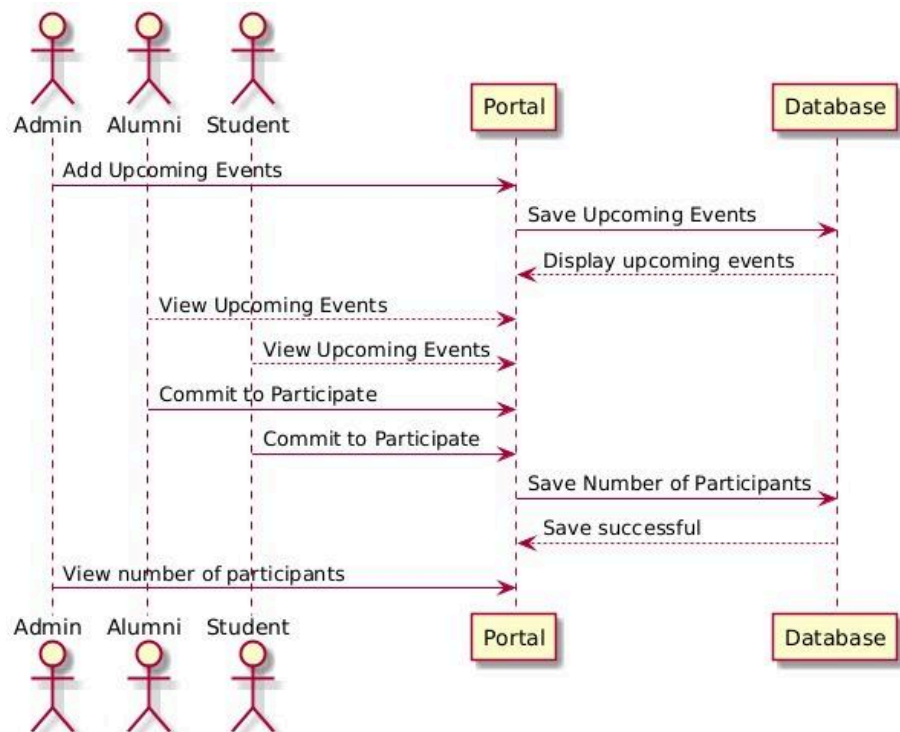
(3) Sequence Diagram



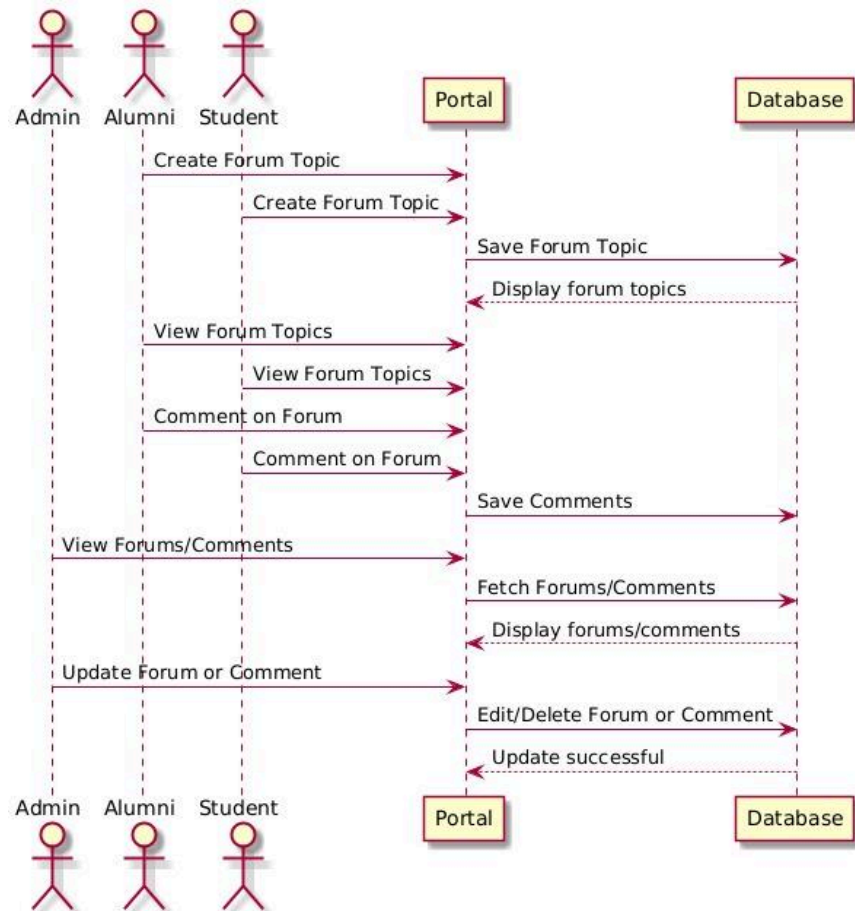
Registration/Login



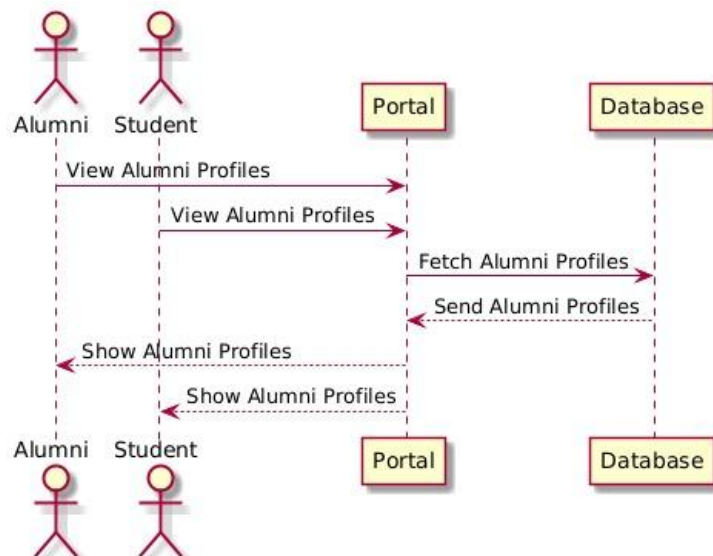
Job/Education Postings



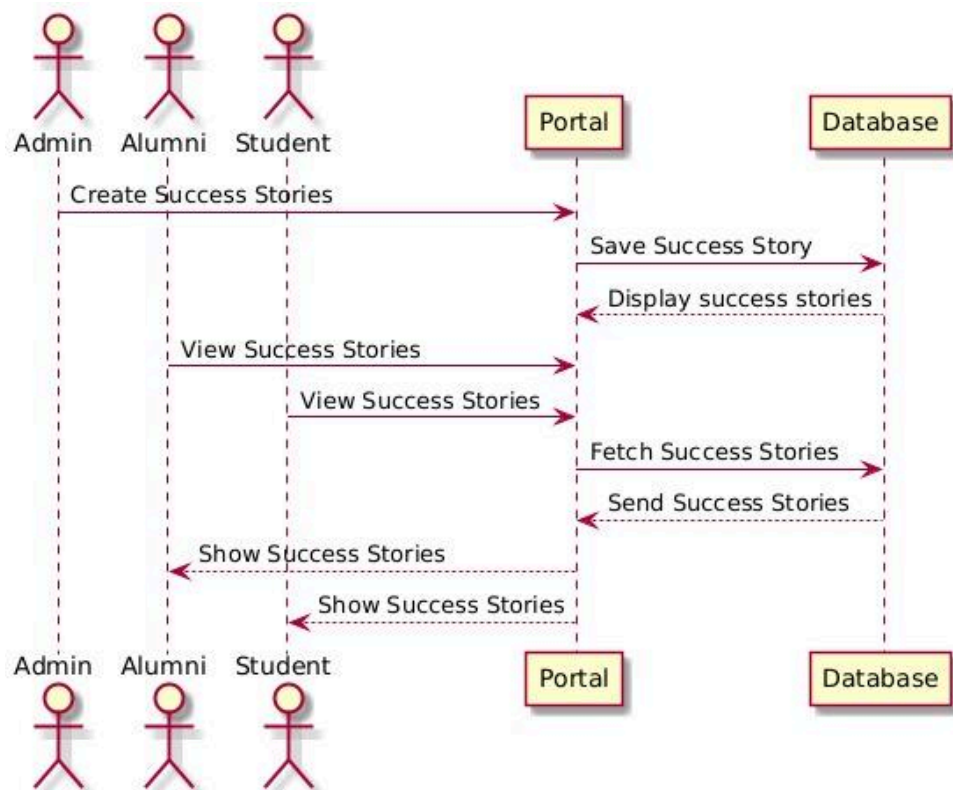
Upcoming Events



Forums

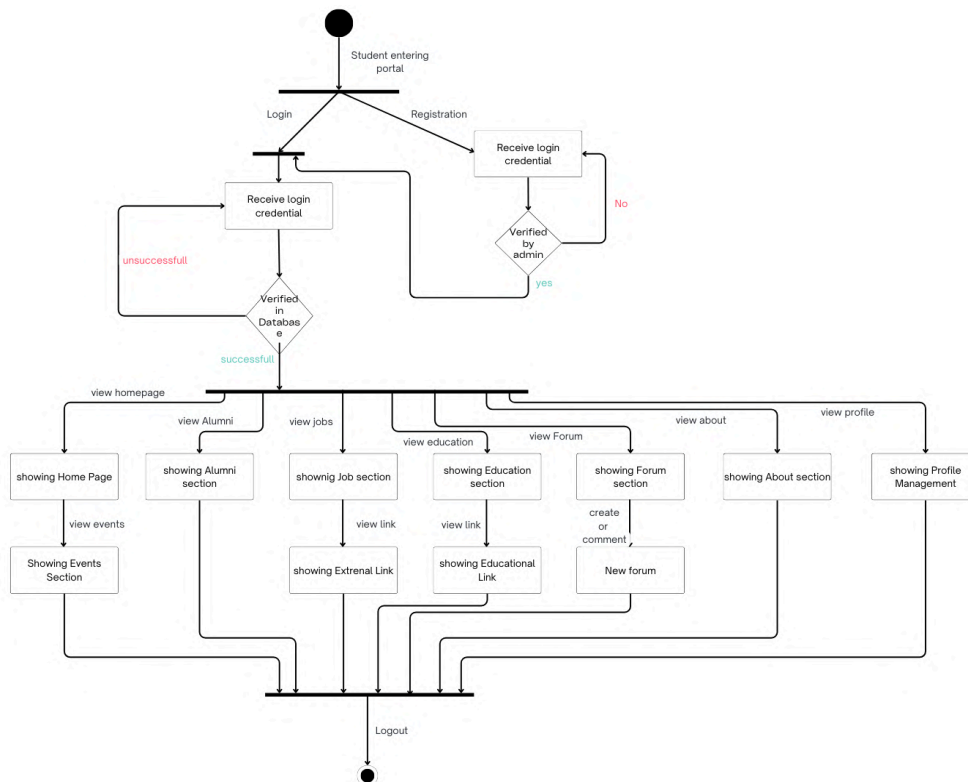


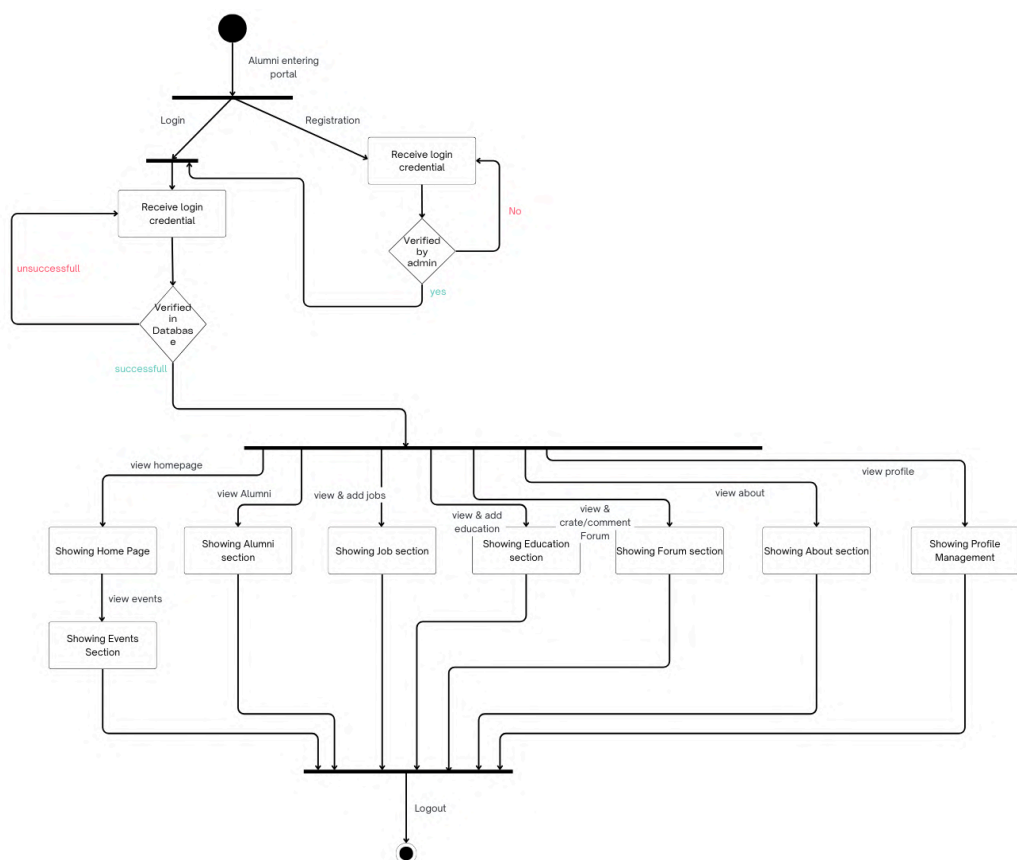
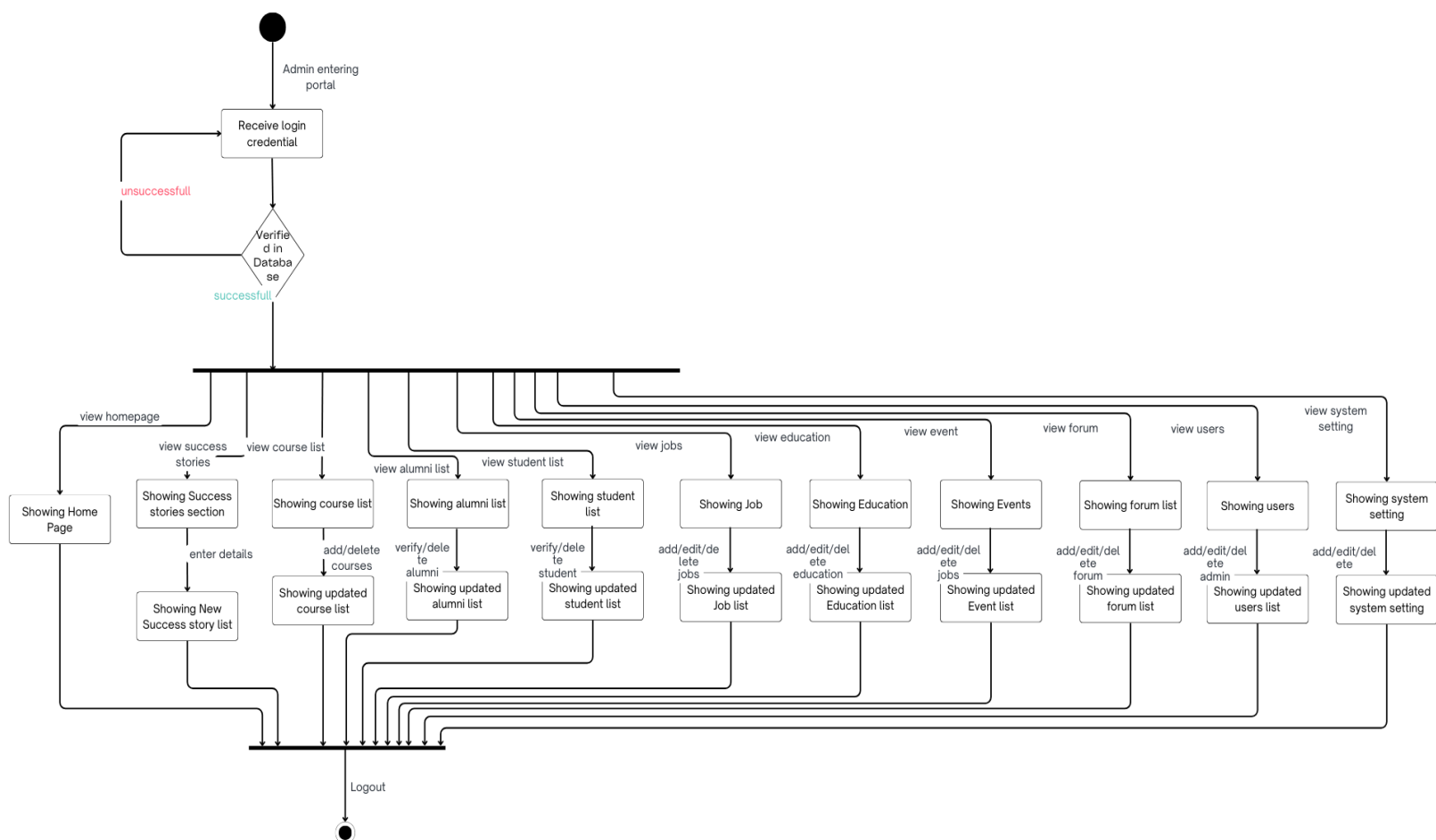
View Alumni Profiles



Success Stories

(4) State Diagram





Activity Diagram

